

POKHARA UNIVERSITY

Level: Bachelor
Programme: BE
Course: Calculus II

Semester: Fall

Year : 2023
Full Marks: 100
Pass Marks: 45
Time : 3hrs.

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Attempt all the questions.

1. a) Evaluate $\int_0^2 \int_0^{\sqrt{4-y^2}} \cos(x^2 + y^2) dx dy$ by changing into polar integration. 5
 - b) Evaluate: $\int_0^1 \int_0^1 \int_0^1 (x^2 + y^2 + z^2) dz dy dx$ 5
 - c) Find the volume in the first octant bounded by the coordinate planes, the cylinder $x^2 + y^2 = 4$ and $z + y = 3$. 5
 2. a) Solve the differential equation $y'' + (1 - x^2)y = 0$ by using power series method. 7
 - b) Define Legendre's equation. Also derive the solution of Legendre's equation. 8
- OR**
- If $J_n(x)$ represents the Bessel's function of order n then show that:
- i. $\frac{d}{dx} [x^n J_n(x)] = x^n J_{n-1}(x)$.
 - ii. $J_{\frac{1}{2}}(x) = \sqrt{\frac{2}{\pi x}} \sin x$.
3. a) State First Shifting Theorem of Laplace transforms. 8
 - i. Find the Laplace transform of $te^{-t} \cosh t$
 - ii. Find the inverse Laplace transform of $\frac{1}{s^2 - 5s + 6}$
 - b) Solve the initial value problem:
 $y'' + 4y' + 3y = e^{-t}$, $y(0) = y'(0) = 1$ by using Laplace transform. 7
 4. a) If $\Phi = \log(x^2 + y^2 + z^2)$ then find $\text{div}(\text{grad}\Phi)$. 5

- b) Find the directional derivative of $f = x^2yz + 4xz^2$ at $(1, -2, 1)$ in the direction $2\vec{i} - \vec{j} - 2\vec{k}$. 5
 - c) Calculate: $\oint_C \vec{F} \cdot d\vec{r}$ if $\vec{F} = (y, z, x)$, $C: \vec{r} = (t, t^2, t^3)$, from $(0, 0, 0)$ and $(2, 4, 8)$. 5
 5. a) Define Green's Theorem. Evaluate $\oint_C \vec{F} \cdot d\vec{r}$, where 8

$\vec{F} = (\sin y, \cos x)$ and C is the triangle with vertices $(0, 0)$, $(\pi, 0)$, $(\pi, 1)$, by using Green's theorem.
 - b) Evaluate $\oint_C \vec{F} \cdot d\vec{r}$ by using Stoke's theorem, where $\vec{F} = (y^2, z^2, x^2)$ 7

and C is the boundary of the surface $S: x + y + z = 1$ in the first octant.
- OR**
- Evaluate the surface integral $\iint_S \vec{F} \cdot \vec{n} dA$ where $\vec{F} = (x^2, y^2, z^2)$ and $S: \vec{r} = (u \cos v, u \sin v, 3v)$, $0 \leq u \leq 1$, $0 \leq v \leq 2\pi$
6. a) Find the fourier series of $f(x) = x + |x|$ for $-\pi < x < \pi$. 7
 - b) Expand the function $f(x) = x^2$ in the interval $0 < x < \pi$ in half range Fourier cosine series and show that $\frac{\pi^2}{6} = 1 + \frac{1}{4} + \frac{1}{9} + \frac{1}{16} + \dots$ 8
 7. Solve Any Two: 2x5
 - a) Find the general solution of $2u_x + 2u_y - u = 0$.
 - b) Derive the one-dimensional traffic flow model using conservation law.
 - c) Show that the value under integral sign $\int_{(4,0,3)}^{(-1,1,2)} [(yz + 1)dx + (xz + 1)dy + (xy + 1)dz]$ is exact and evaluate the integral.

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Attempt all the questions.

1. a) Evaluate the integral by reversing the order of integration: 5

$$\int_0^{\pi} \int_x^{\pi} \left(\frac{\sin y}{y} \right) dy dx$$

- b) Evaluate the integral: $\int_0^1 \int_0^{1-x} \int_0^{x+y} e^z dz dy dx$ 5

- c) Find the volume of the solid whose base is the region in the xy -plane that is bounded by the parabola $y = 4 - x^2$ and the line $y = 3x$, while the top of the solid is bounded by the plane $z = x + 4$. 5

2. a) Solve by using power series method: $y'' - 4xy' + (4x^2 - 2)y = 0$. 7

- b) Solve the Bessel's equation $x^2 y'' + xy' + (x^2 - n^2)y = 0$. 8

OR

- i. Express $x^2 - 4x + 2$ as a Legendre polynomial by using Rodrigues formula.

- ii. Sketch the graph of $P_0(x)$, $P_1(x)$, $P_2(x)$ and $P_3(x)$ in a single graph.

3. a) Solve the initial value problem by using Laplace transform: 7

$$y'' + 2y' + 2y = 5 \sin t, y(0) = y'(0) = 0$$

- b) i. Find the Laplace transform of: $\cosh at \sin at$. 8

- ii. Find the inverse Laplace transform of $\frac{s}{(s^2+1)(s^2+4)}$ using the convolution theorem.

4. a) If $\phi = x^3 + y^3 + z^3 - 3xyz$, find: $\text{div}(\text{grad} \phi)$ 5

- b) Find the directional derivative of $\phi = x^2 - y^2 + 2z^2$ at the point $P(1, 2, 3)$ in the direction of line PQ, where Q is the point $(5, 0, 4)$. 5

- c) Find the work done by the force $\vec{F} = (2y+3)\vec{i} + xz\vec{j} + (yz-x)\vec{k}$ when a particle moves from $(0, 0, 0)$ to the point $(2, 1, 1)$ along the curve $x = 2t^2, y = t, z = t^3$. 5

5. a) Find the flux of $\vec{F} = z\vec{i} + x\vec{j} + y\vec{k}$ over the portion of the surface $x^2 + y^2 = 1$ in the first octant between $z = 0$ to $z = 2$. 7
- b) State Stoke's theorem. Use it to evaluate $\oint_C \vec{F} \cdot d\vec{r}$, where $\vec{F} = (z, x, y)$ and C is the surface of upper half of sphere $x^2 + y^2 + z^2 = a^2$ in xy -plane. 8

OR

State Gauss divergence theorem. Evaluate $\iiint_S \vec{F} \cdot \vec{n} ds$, where $\vec{F} = (4x, x^2y, -x^2z)$ and S is the surface of the tetrahedron with vertices $(0,0,0)$, $(1,0,0)$, $(0,1,0)$, and $(0,0,1)$, by using the Gauss divergence theorem.

6. a) Find the Fourier series of the function $f(x) = |x|$ for $-2 < x < 2$. 7

- b) Expand the function $f(x) = x^2$ in the interval $0 < x < \pi$ in half-range Fourier cosine series and deduce that $\frac{\pi^2}{6} = 1 + \frac{1}{2^2} + \frac{1}{3^2} + \frac{1}{4^2} + \dots$ 8

7. Solve: (Any two) 2x5

- a) Find the general solution of $u_x + u_y - u = 0$.

- b) Derive one dimensional traffic flow model using convection law.

- c) Show that the integral $\int_{(0,0,0)}^{(2,4,0)} e^{x-y+z^2} (dx - dy + 2zdz)$ is exact and evaluate the exact value.

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Attempt all the questions.

1. a) Evaluate: $\int_0^\pi \int_x^\pi \frac{\sin y}{y} dy dx$ by changing the order of integration. 5
 - b) Evaluate the integrals: $\iiint_V (x^2 + y^2 + z^2) dx dy dz$ taken over the volume of the sphere $x^2 + y^2 + z^2 = 1$. 5
 - c) Find the volume of the solid whose base is the region in the xy-plane that is bounded by the parabola $y = 4 - x^2$ and the line $y = 3x$, while the top of the solid is bounded by the plane $z = x + 4$. 5
 2. a) Solve the differential equation: $y'' + 9y = 0$ by using power series method. 7
 - b) Define Legendre differential equation. Also, derive the solution of Legendre differential equation. 8
- OR**
- If $J_n(x)$ represents the Bessel's function of order n , then show that:
- i. $\frac{d}{dx} [x^n J_n(x)] = x^n J_{n-1}(x)$ 4
 - ii. $J_{\frac{1}{2}}(x) = \sqrt{\frac{2}{\pi x}} \sin x$ 4
3. a) Solve the following differential equation by using Laplace transform $y'' + 2y' - 3y = 6e^{-2t}$, $y(0) = 2$, $y'(0) = -14$. 7
 - b) i) State second shifting property for Laplace transform and using this, find the Laplace transform of $e^{-3t} u(t-3)$. 4
 - ii) Find the inverse Laplace transform of $\frac{\pi}{(s-3)^2 + \pi^2}$. 4
 4. a) Find the directional derivative of $f = x^2 yz + 4xz^2$ at $(1, -2, 1)$ in the direction $2\vec{i} - \vec{j} - 2\vec{k}$. 5

- b) Show that the vector $\vec{F} = (x^2 - yz, y^2 - xz, z^2 - xy)$ is conservative vector field and find the scalar potential function ϕ such that $\vec{F} = \nabla \phi$. 5
 - c) Find the work done by the force $\vec{F} = (3x^2, 2xz - y, z)$ along the curve $x = t$, $y = \frac{t^2}{4}$, $z = \frac{3t^2}{8}$ from $(0, 0, 0)$ to $(2, 1, 3)$. 5
 5. a) Evaluate the integral by using Green's theorem 7
- $\oint_C [(3x^2 + y) dx + 4y^2 dy]$ where C : the boundary of the triangle with vertices $(0, 0)$, $(1, 0)$, $(0, 2)$; oriented in counterclockwise direction.
- b) State Stoke's Theorem. Evaluate $\oint_C (\vec{F} \cdot d\vec{r})$ by using Stoke's theorem, where $\vec{F} = (y, xz^3, -zy^3)$ and $C: x^2 + y^2 = 4$, $z = 3$. 8
- OR**
- Evaluate: $\iint_S \vec{F} \cdot \vec{n} dA$, where $\vec{F} = (18z, -12, 3y)$ and S is the surface of plane $2x + 3y + 6z = 12$ in the first octant.
6. a) Find the Fourier series of $f(x) = x + |x|$, $-\pi < x < \pi$ and show that $\frac{\pi^2}{8} = 1 + \frac{1}{9} + \frac{1}{25} + \dots$ 7
 - b) Find the Fourier sine and cosine series for the function $f(x) = x^2$ in the interval $0 < x < L$. 8
 7. Write short notes on: (Any two) 2x5
- a) Find the general solution of the partial differential equation $u_x + u_y = u$.
 - b) Derive the one-dimensional traffic flow model using conservation law.
 - c) A particle moves along the curve $x = t^3 + 1$, $y = t^2$, $z = 2t + 5$. Find the velocity and acceleration at $t = 1$.

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Attempt all the questions.

1. 5
 - a) Evaluate $\int_0^2 \int_0^{\sqrt{4-y^2}} \cos(x^2 + y^2) dx dy$ by changing into polar integration. 5
 - b) Evaluate: $\int_0^1 \int_0^1 \int_0^1 (x^2 + y^2 + z^2) dz dy dx$ 5
 - c) Find the volume in the first octant bounded by the coordinate planes, the cylinder $x^2 + y^2 = 4$ and $z + y = 3$. 5
2. 8
 - a) i) Find the Laplace Transform of: $\frac{\sin^2 t}{t}$ 8
 ii) Find the inverse Laplace Transform of: $\frac{w}{s^2(s^2 + w^2)}$
 - b) Solve the initial value problem: $y'' + 2y' - 3y = 6e^{-2t}$ where, $y(0) = 2$, $y'(0) = -14$ using Laplace Transform. 7
3. 7
 - a) Solve $y'' + y = 0$ by using power series method. 7
 - b) Derive Rodrigue's formula for Legendre polynomials. 8

OR

Define Bessel's differential equation of order n . Find its complete solution.
4. 5
 - a) Find the directional derivatives of $f = xy^2 + yz^3$ at $P(2, -1, 1)$ along the direction of the normal to the surface $x \log z - y^2 + 4 = 0$ at $(-1, 2, 1)$. 5
 - b) If $\phi = \log(x^2 + y^2 + z^2)$, find $\text{div}(\text{grad } \phi)$ and $\text{curl}(\text{grad } \phi)$. 5
 - c) Show that the value under integral sign is exact and then evaluate integral, 5

- $(1, 1, 1)$
 $\int [yz \sinh(xz) dx + \cosh(xz) dy + xy \sinh(xz) dz]$
 $(0, 2, 3)$
5. a) Let S be the hemi-sphere given by $z = \sqrt{9 - x^2 - y^2}$ and $\vec{F} = (3x, 3y, z)$. Find the flux of $\vec{F}$ through S .
 - b) State Gauss divergence theorem. Evaluate $\iint_S \vec{F} \cdot \vec{n} dA$ where $\vec{F} = (y^3, x^3, z^3)$ and S is the cylinder $x^2 + 4y^2 = 1$ $0 < z < 5$, $x > 0$, $y > 0$ by using it.

OR

- State Stoke's theorem. Evaluate $\oint_C \vec{F} \cdot d\vec{r}$ where $\vec{F} = (y^3, 0, x^3)$ and C is the boundary of the triangle with vertices are given by the vectors $(1, 0, 0)$, $(0, 1, 0)$, $(0, 0, 1)$.
6. a) Find the fourier series of $f(x) = x + |x|$ for $-\pi < x < \pi$.
 - b) Find the fourier cosine series of $f(x) = x \sin x$, $0 < x < \pi$ then show that $\frac{\pi-2}{4} = \frac{1}{1.3} - \frac{1}{3.5} + \frac{1}{5.7} - \frac{1}{7.9} + \dots$
 7. Attempt any two questions. 2
 - a) Solve: $u_t + 2uu_x = 0$, $u(x, 0) = e^{-x}$.
 - b) Derive the one-dimensional traffic flow model using conservation law.
 - c) Find the angle between the surfaces $xy^2z - 3x - z^2 = 0$ and $3x^2 + 2z = 1 + y^2$ at $(1, -2, 1)$